

## ? The Question

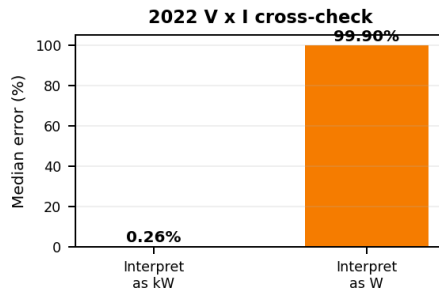
When throughput falls 78%, what does the evidence support about data quality, recorded commands, and battery health?

M5BAT Pb1 | flooded lead-acid | 2017-2025 at 1 s  
First and last years are partial; usable coverage varies.

## What the Data Shows

- 1 Power is numerically kW-scale although source metadata labels it W.
- 2 2022->2024: actual energy -78.0%; recorded command energy -77.4%.
- 3 Filtered DCIR: 86.8 -> 99.9 mOhm (+15.1%); not distribution-matched.

### 1 Signal Trust



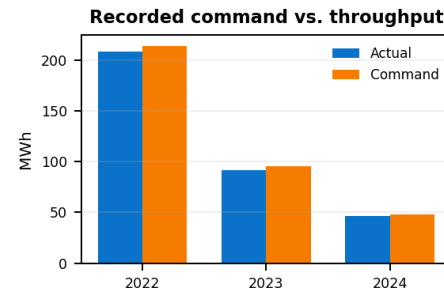
V x I reproduces the power scale; interpreting raw power as W gives ~99.9% median error.

#### Counter excerpt:

703,972 -> 3,185,836,042 -> 703,972

Exact 16-bit word-swap arithmetic; public source locator is partial.

### 2 Recorded Command vs. Throughput



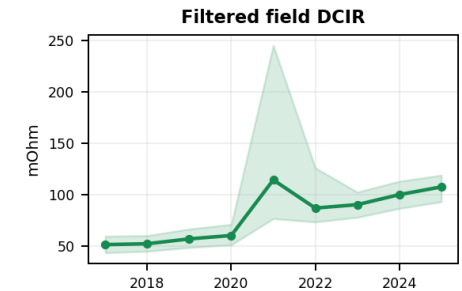
Lower energy accompanies lower recorded commands.  
Throughput alone does not establish proportional capacity loss.

#### Important:

This does not explain why upstream commands declined.

Actual/command also falls ~97.45% -> 95.22%.

### 3 Filtered Field DCIR



2022->2024 median: 86.8 -> 99.9 mOhm (+15.1%).  
2021 unresolved discontinuity; 2025 is a partial year.

#### Boundary:

Common-condition filtered, not distribution-matched or HPCC.

P25/P75 are event dispersion, not confidence intervals.

## i Engineering Context

- > **Measurement boundary**  
BSC DC V/I are measured at the inverter DC input; temperature is cabinet air, not cell temperature.
- > **PQ-mode**  
The source PQ flag is FCR-related but not FCR-exclusive.
- > **Coverage**  
Pre-2022 BMS/BSC usable coverage differs; coverage is not hardware availability.
- > **Counter evidence**  
Published word-swap excerpts have partial public source locators.

## ! Engineering Interpretation

- > **Observed envelope**  
Tracking evidence covers only the selected slow-ramp envelope; it does not rule out upstream derating.
- > **Sparse high power**  
450-700 kW discharge support is only ~27 s / ~26 s in 2023 / 2024; do not generalize.
- > **AC/DC discrepancy**  
AC/DC > 1 prevents direct efficiency or battery-loss attribution.
- > **Health signal**  
BSC-side apparent string-path resistance rises; this is not isolated cell resistance or a measured capacity-loss rate.

## > Decision Relevance

- > **Validate data first**  
Do not base long-term decisions on mislabeled power or unvalidated cumulative counters.
- > **Review limits next**  
Inspect supervisory commands, derating, SOC protection and availability logic before calling capacity loss.
- > **Verify health separately**  
Retain a resistance concern and confirm it with better-matched or controlled health evidence.
- > **Evidence-led prioritization**  
This public-data case demonstrates diagnostic prioritization, not a verified site recovery or financial saving.



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Before deciding what is wrong,  
determine whether the data supports  
the conclusion.